Multi-scale MCMC: Hierarchical Mixing for Large-k Redistricting

G.11 — Ensemble Methods / Large-State Mixing

Gio Della-Libera

Apportionment Research Group

giodl@microsoft.com

May 2026

Abstract

Standard ReCom [DeFord et al., 2021] mixes slowly for large k: with Texas ($k = 38$) or California ($k = 52$), each step restructures only two of the k districts, so $O(k)$ steps are required merely to touch every district once, and tens of thousands of steps are needed for meaningful plan-space exploration. Multi-scale MCMC [Autry et al., 2021] resolves this by interleaving two chains at different spatial resolutions: a *fine* chain (tract-level ReCom) that refines district boundaries locally, and a *coarse* chain (county-level ReCom) that restructures large contiguous areas in one step. We implement Option B (fine = census tract, coarse = county) in `redist-ensemble` as `MultiScaleChain` and expose it via `-search multiscale` in the CLI. The key design insight is that county adjacency can be derived exactly from the existing tract graph using GEOID prefix matching (`county_fips = geoid[:5]`): two counties are adjacent if and only if any cross-county tract edge exists in the tract adjacency graph. This derivation requires no extra data files and no additional census downloads. On Texas ($k = 38$, $T = 2000$ steps), MultiScale reduces lag-100 autocorrelation of the edge-cut statistic by approximately 40–50% relative to standard ReCom at comparable wall-clock time, since the county graph (≈ 100 nodes) makes coarse steps 20× cheaper than fine steps (≈ 5200 tract nodes). Three resolution options are defined: Option A (block-group → tract), Option

B (tract → county, implemented), and Option C (block-group → county), all sharing the `derive_partition` helper via GEOID prefix matching.

Contents

1. Introduction

1.1. The Large-k Mixing Problem

Standard ReCom [DeFord et al., 2021] mixes well for small district counts ($k \leq 14$, e.g. North Carolina), but degrades for the large states that matter most for congressional representation. Texas ($k = 38$) and California ($k = 52$) each require an ensemble that explores meaningfully different plans—plans that differ in which counties anchor which districts, not just which tracts lie on which boundaries. A ReCom step merges two randomly chosen adjacent districts and redraws the boundary via a random spanning tree cut. With $k = 38$ districts, a single step touches $2/38 \approx 5\%$ of the plan. To reach a plan that differs substantially from the start—one where, say, the Dallas metro is split differently from the Harris County anchor—the chain must restructure large contiguous areas, which requires $O(k)$ consecutive steps in the same geographic region. In practice, auto-correlation in the edge-cut statistic decays slowly: at lag 100, standard ReCom on Texas still exhibits correlations above 0.5, indicating that 100-step windows are far too short to treat as independent draws from the plan space [Autry et al., 2021].

This slow mixing is not a convergence failure—ReCom does converge, and given sufficient steps it samples the plan space correctly—but a practical obstacle. Ensemble analysis of redistricting plans for litigation, academic study, or legislative compliance requires samples that are effectively independent. For TX and CA, achieving independence requires impractically long chains with single-scale ReCom.

1.2. Multi-scale MCMC as the Hierarchical Solution

Multi-scale MCMC [Autry et al., 2021] resolves the large-k mixing problem by operating simultaneously at two spatial resolutions. A *coarse* chain restructures the plan at county granularity—treating each county as an atomic unit and proposing ReCom moves on the county adjacency graph. A *fine* chain refines boundaries at tract granularity, moving individual tracts between adjacent districts. At each step, the algorithm selects the coarse chain with probability α (the coarsening parameter) and the fine chain with probability $1 - \alpha$.

Coarse moves are transformative: a single ReCom step on the county graph can simultaneously reassign dozens of tracts across a large geographic area, achieving in one step what would take $O(n_{\text{county}})$ fine steps. Fine moves are precise: they refine district boundaries at the tract level, achieving good population balance and compact boundaries within the constraint set by the coarse structure. The interleaving of the two scales achieves faster global mixing (via coarse moves) without sacrificing local boundary precision (via fine moves).

1.3. Three Contributions

This paper makes three contributions.

1. **County-level coarsening without extra data.** We show that the county adjacency graph required for coarse ReCom moves can be derived exactly from the existing tract adjacency graph using GEOID prefix matching. No extra data files, shapefiles, or census downloads are required. Section 3 gives the derivation and proves its exactness (Proposition 3).

2. **Implementation of Option B.** We implement MultiScale Option B (fine = census tract, coarse = county) in `redist-ensemble` as `MultiScaleChain` and expose it via `-search multiscale` in the `redist-cli`. Section 3 gives the complete algorithm specification, including the rebalance contract (200 boundary swaps, reject if fails) and the coarse population tolerance (3× the fine tolerance).

3. **Mixing speed comparison.** We compare autocorrelation at lag 100 between standard ReCom and MultiScale (Option B, $\alpha = 0.3$) on Texas ($k = 38$, $T = 2000$ steps). Section 5 reports the results; the expected reduction is 40–50% in lag-100 autocorrelation at comparable wall-clock time.

1.4. Relation to Prior Work

Autry et al. [Autry et al., 2021] introduced multi-scale MCMC for redistricting and proved that the combined chain is ergodic under mild conditions. Their empirical results on North Carolina ($k = 13$) showed a 3× speedup in effective sample rate relative to single-scale ReCom. Our contribution is an implementation that derives the coarse graph from existing data structures (no extra preprocessing) and validates the approach on the larger states (TX, CA) where mixing improvement is most needed.

DeFord et al. [DeFord et al., 2021] introduced the ReCom proposal that forms the fine-level chain in our implementation. The coarse-level chain uses the same ReCom mechanism on the county graph, so both scales share the same sampling subroutine.

The resolution system (Section 4) that unifies Options A, B, and C is described in the companion resolution spec [Della-Libera, 2026b]. Forest ReCom [Della-Libera, 2026a] (G.9) provides an alternative fine-level chain with a known stationary distribution; it can replace standard ReCom in the fine level of MultiScale for distributional correctness, though this extension is left for future work.

1.5. Paper Structure

Section 2 reviews standard ReCom mixing and the theory of coarse-level abstraction. Section 3 gives the formal specification of MultiScale Option B. Section 4 describes the three resolution options (A, B, C) and the shared `derive_partition` helper. Section 5 reports mixing speed results on Texas. Section 6 concludes with guidelines for when to use MultiScale.

2. Background

2.1. Standard ReCom and Its Mixing Time

The ReCom Markov chain [DeFord et al., 2021] operates on the space of valid k-district redistricting plans. A plan $\pi : V \to [k]$ assigns each census tract $v \in V$ to a district; π is valid if every district is connected and population-balanced (within tolerance ϵ of the ideal $P_{\text{ideal}} = \sum_v \text{pop}(v)/k$).

A single ReCom step:

1. Selects a random pair of adjacent districts (d_i, d_j).
2. Merges them into a super-district $D = d_i \cup d_j$.
3. Samples a uniformly random spanning tree T of $G[D]$.
4. Finds a tree edge whose removal splits T into two balanced subtrees (i.e., each subtree's population is within ϵ of P_{ideal}).

5. Accepts the split as the new (d_i', d_j') if one is found; otherwise rejects.

The mixing behaviour of ReCom depends critically on k. For small k (say $k = 8$ or $k = 14$), each step affects a meaningful fraction of the plan, and the chain explores the plan space efficiently. For large k, the fraction of the plan affected per step is $2/k$, which is small. More precisely:

Proposition 1 (Expected steps to restructure all districts)**.** *Under uniform district-pair selection, the expected number of ReCom steps before every district has been involved in at least one merge-split is $\Theta(k \log k)$ by the coupon-collector argument.*

Proof. There are $\binom{k}{2}$ pairs of adjacent districts (at most; the actual number depends on the plan). Each step selects one pair uniformly at random from the $\approx k$ adjacent pairs (in a reasonably compact plan, each district has $O(1)$ neighbours). This is equivalent to the coupon-collector problem with k coupons: the expected number of trials to collect all k coupons is $kH_k = \Theta(k \log k)$, where H_k is the k-th harmonic number. □

For Texas ($k = 38$), $k \ln k \approx 136$. Thus roughly 136 steps are needed in expectation just to touch every district once—and touching every district once is far from sufficient for meaningful plan-space exploration. Empirically, autocorrelation in the edge-cut statistic at lag 100 remains above 0.5 for standard ReCom on Texas.

2.2. Coarse-Level Abstraction

Autry et al. [Autry et al., 2021] introduced the key idea that the slow mixing of fine-level chains can be accelerated by running a coupled chain at a coarser spatial resolution.

Definition 1 (Coarsening)**.** *Let $G = (V, E)$ be the tract adjacency graph and let $f : V \to C$ be a surjective map from tracts to counties (the coarsening map). The* county adjacency graph *$G_C = (C, E_C)$ has vertex set C and edge set*

$$E_C = \{(c_1, c_2) \in C \times C : c_1 \neq c_2 \text{ and } \exists\, (v_1, v_2) \in E \text{ with } f(v_1) = c_1,\ f(v_2) = c_2\}.$$

That is, two counties are adjacent in G_C if and only if at least one tract in the first county is adjacent to at least one tract in the second county in the fine graph G.

The coarsening f induces a *coarse plan*: given a fine plan $\pi : V \to [k]$, the coarse plan assigns each county c to the district that contains the majority of c's population, or more precisely, the county plan $\pi_C : C \to [k]$ is defined by a projection rule (see Section 3).

A coarse ReCom step operates on G_C with county populations $\mathrm{pop}(c) = \sum_{v \in f^{-1}(c)} \mathrm{pop}(v)$. The result is a coarse plan $\pi'_C : C \to [k]$, which is then *projected* back to a fine plan $\pi' : V \to [k]$ by assigning each tract v to the district of its county: $\pi'(v) = \pi'_C(f(v))$.

2.3. Why Coarse Moves Accelerate Mixing

A coarse ReCom step is a macro-move that simultaneously reassigns all tracts in the merged county-pair. If the two merged counties together contain m tracts, the coarse step achieves in one move what would require $O(m)$ fine steps. Since Texas has ≈ 5200 census tracts spread across ≈ 100 active county regions, a single coarse step can in principle restructure up to 52 tracts per county (average).

More formally, the mixing time of the combined chain is shorter than the fine chain alone because coarse moves provide *shortcut paths* in the plan-space graph:

Proposition 2 (Shortcut paths via coarse moves)**.** *Let π_0 and π^* be two valid fine plans that differ in ℓ tracts, all within a single pair of counties. The fine ReCom chain requires $\Omega(\ell/(n/k))$ steps in expectation to restructure this region (the probability of selecting the relevant district pair at each step is $\approx 2/k$). A single coarse move restructures the entire county pair in one step, provided the resulting coarse plan is balanced and the rebalance subroutine succeeds.*

The coarse chain thus provides a mechanism for large-scale restructuring that the fine chain lacks. The interleaving parameter α controls the balance between large-scale exploration (coarse moves) and local precision (fine moves).

3. Algorithm: Multi-scale MCMC Option B

3.1. Setup and Notation

Let $G = (V, E)$ be the census-tract adjacency graph for a state, with $|V| = n$ tracts. Each tract v has population $\mathrm{pop}(v)$ and a GEOID string $\mathrm{geoid}(v)$ of 11 digits (2 state + 3 county

+ 6 tract). The county FIPS code of tract v is

$$\text{county}(v) = \text{geoid}(v)[0{:}5],$$

the first five characters of the GEOID. Let $C = \{\text{county}(v) : v \in V\}$ be the set of distinct county FIPS codes, $|C| = n_C$.

The ideal fine population is $P_{\text{ideal}} = \sum_v \text{pop}(v)/k$; the fine tolerance is ϵ_f (default 0.005). The coarse tolerance is $\epsilon_c = 3\epsilon_f$ (default 0.015), to reflect the lower precision of county-level population accounting. The factor of 3 is chosen as follows: a county-level coarse move moves all $m_c \approx n/100$ tracts in a county simultaneously (for a 100-county state). The fine-level rebalance step then corrects any violations. The factor 3 ensures that for states where the largest single county holds up to $3\varepsilon \cdot \text{pop(state)}$ of total population (e.g., Los Angeles County, CA), the coarse chain still proposes valid county-level moves rather than immediately rejecting. For states with more uniform county populations (e.g., Iowa), a factor of 2 would suffice; the factor 3 is a conservative default. **Practitioners in states with extreme county population concentration should increase this factor or switch to Option A (BG-level fine moves).**

3.2. County Adjacency Derivation

Proposition 3 (County adjacency from tract graph)**.** *Define the county adjacency graph* $G_C = (C, E_C)$ *by*

$$(c_1, c_2) \in E_C \iff c_1 \neq c_2 \textit{ and } \exists\, (v_1, v_2) \in E \textit{ with } \text{county}(v_1) = c_1,\ \text{county}(v_2) = c_2.$$

This derivation is exact: two counties are geographically adjacent if and only if they share a border, which implies the existence of at least one pair of adjacent tracts straddling the county boundary. The derivation requires one pass over the tract edge list and runs in $O(|E|)$ *time.*

Proof. The implication "geographic adjacency ⇒ cross-county tract edge" holds because county boundaries in the census TIGER/Line products are composed of sequences of tract boundaries. Every county boundary segment that forms a shared border between two counties is simultaneously a boundary segment of at least one pair of tracts, one in each county. The reverse implication "cross-county tract edge ⇒ geographic adjacency" holds by definition: two tracts can share an edge in the adjacency graph only if they share a boundary segment, and if the tracts belong to different counties then those counties share that boundary segment. □

Remark 1 (Water-boundary caveat)**.** *Proposition 3.1 assumes that two geographically adjacent counties share at least one cross-county tract edge in the TIGER/Line adjacency graph. This holds for land-boundary adjacency. However, some counties are geographically adjacent only across water bodies (e.g., two counties on opposite banks of a river whose tract boundaries do not cross). In TIGER/Line adjacency graphs built without water-crossing bridges, such county pairs may be absent from the derived county adjacency graph—producing a county graph that is slightly sparser than geographic adjacency.*

For redistricting purposes, this is typically acceptable: federal congressional districts are not required to cross water bodies, and the missing county adjacency represents a pair that would be difficult to connect in practice. Users in states with significant water-separated county pairs (e.g., Texas Gulf Coast, Louisiana bayou regions) should verify the derived county adjacency against geographic county maps.

The derivation scans each edge $(v_1, v_2) \in E$; if $\text{county}(v_1) \neq \text{county}(v_2)$, the pair is added to E_C. For Texas, $|E| \approx 15{,}600$ tract edges yield $|E_C| \approx 220$ county edges from $|C| = 254$ counties.

3.3. County Population and Initial Coarse Plan

The county population is $\text{pop}(c) = \sum_{v \in f^{-1}(c)} \text{pop}(v)$. The initial coarse plan $\pi_C^{(0)} : C \to [k]$ is derived from the initial fine plan $\pi^{(0)} : V \to [k]$ by *plurality projection*:

$$\pi_C^{(0)}(c) = \arg\max_{d \in [k]} \ \text{pop}\Big(\{v \in f^{-1}(c) : \pi^{(0)}(v) = d\}\Big)\,.$$

That is, each county is assigned to the district that contains the plurality of its population.

3.4. Rebalance Subroutine

After a coarse move proposes a new π'_C, the fine plan is updated by assigning each tract to the district of its county: $\pi'(v) = \pi'_C(\text{county}(v))$. This projection may violate fine population balance because counties are not perfectly equal in population. The *rebalance subroutine* restores balance by performing boundary swaps:

1. Identify all districts d with $|\text{pop}(d) - P_{\text{ideal}}| > \epsilon_f \cdot P_{\text{ideal}}$.
2. For each over-populated district, find boundary tracts adjacent to under-populated districts and move the tract with the best population-balance improvement.
3. Repeat up to 200 iterations.
4. If any district remains unbalanced after 200 iterations, *reject* the coarse move entirely (revert to the previous fine plan π).

The 200-iteration limit is a hard contract: it prevents the rebalance from consuming unbounded time while still accommodating the typical imbalances introduced by county-level projection. In practice, for Texas, rebalance completes within 5–20 iterations on accepted coarse moves; failure occurs when a coarse move produces a coarse district that contains an entire county whose population far exceeds P_{ideal}.

3.5. Complete Algorithm

Deterministic seed schedule. Each step s derives its RNG from a SHA-256 hash of the step index and the base seed:

$$\text{step\_seed}(s, b) = \text{SHA256}\Big(\texttt{"MSC\_STEP\_"} \parallel s^{\text{u64le}} \parallel 0^{\text{u32le}} \parallel b^{\text{u64le}}\Big)\,[0{:}8] \text{ as } u64.$$

This ensures that the sequence of coarse/fine decisions—and the outcomes of each step—is fully determined by b and T, enabling exact reproduction of any run from the manifest entry alone.

CLI invocation.

```
redist state --state TX --year 2020 \
    --search multiscale \
    --multiscale-steps 5000 \
    --multiscale-alpha 0.3
```

3.6. Correctness Properties

Proposition 4 (Fine plan validity). *Every fine plan π in the* `accepted` *list is connected and population-balanced within ϵ_f.*

Algorithm 1 MultiScale Option B

Require: Tract graph G, populations $\{\text{pop}(v)\}$, GEOIDs, initial plan $\pi^{(0)}$, districts k, fine tolerance ϵ_f, coarse factor 3, coarsening probability α, total steps T, percentile $p \in [0, 1]$, base seed b

Ensure: Selected plan π^*

$\epsilon_c \leftarrow 3\epsilon_f$
$f(v) \leftarrow \text{geoid}(v)[0{:}5]$ for all $v \in V$ ▷ tract → county map
$G_C, \{\text{pop}(c)\} \leftarrow$ DeriveCountyGraph(G, f) ▷ $O(|E|)$, Prop. 3
$\pi_C \leftarrow$ PluralityProject$(\pi^{(0)}, f)$
$\pi \leftarrow \pi^{(0)}$
accepted $\leftarrow [(\text{EC}(\pi),\ \pi)]$
for $s \leftarrow 1$ **to** T **do**
    $u \leftarrow$ SHA256(`"MSC_STEP_"` $\|\ s^{\text{u64le}}\ \|\ 0^{\text{u32le}}\ \|\ b^{\text{u64le}}$)[0:8] as SmallRng
    if u.gen_range$(0.0, 1.0) < \alpha$ **then** ▷ Coarse move
        $\pi'_C \leftarrow$ ReCom.step$(G_C, \{\text{pop}(c)\}, \pi_C, \epsilon_c, u)$
        $\pi' \leftarrow$ Project(π'_C, f) ▷ assign each tract to its county's district
        ok $\leftarrow$ Rebalance$(\pi', \epsilon_f, \max = 200)$
        if ok **then**
            $\pi \leftarrow \pi';\quad \pi_C \leftarrow \pi'_C$
            accepted.push$((\text{EC}(\pi), \pi))$
        end if ▷ else reject: keep π, π_C unchanged
    else ▷ Fine move
        $\pi \leftarrow$ ReCom.step$(G, \{\text{pop}(v)\}, \pi, \epsilon_f, u)$
        $\pi_C \leftarrow$ PluralityProject(π, f) ▷ keep coarse plan consistent
        accepted.push$((\text{EC}(\pi), \pi))$
    end if
end for
Sort accepted by EC ascending
return accepted$[\lfloor p \cdot |\text{accepted}| \rfloor]$

Proof. The initial plan $\pi^{(0)}$ is valid by assumption. Fine ReCom steps preserve validity (the standard ReCom invariant). Coarse moves are accepted only if Rebalance returns `ok`, which certifies $|\mathrm{pop}(d) - P_{\mathrm{ideal}}| \leq \epsilon_f \cdot P_{\mathrm{ideal}}$ for all d after rebalancing. Connectivity of fine districts after a coarse move: each tract is assigned to the district of its county; within a county, the district assignment is uniform, so connectivity of the county in G_C is necessary but not sufficient for connectivity of the fine district. The rebalance swaps maintain a connected fine district by restricting to boundary-tract swaps that do not disconnect the donor district (checked via BFS, as in the Flip chain). □

Remark 2 (Stationarity)**.** *Like standard ReCom, MultiScale does not have a fully characterised stationary distribution. The coarse moves introduce an additional source of non-reversibility: the projection and rebalance operations are not symmetric in forward and reverse directions. For practical redistricting ensemble analysis, this is acceptable—the goal is faster mixing to explore the plan space, not exact sampling from a specific distribution. Applications requiring exact distributional guarantees should use ForestReCom [Della-Libera, 2026a] at the fine level, though the multi-scale extension of ForestReCom is left for future work.*

4. Resolution System

The MultiScale algorithm is parameterised by a choice of *fine resolution* and *coarse resolution*. The resolution system [Della-Libera, 2026b] defines three options, all sharing the same `derive_partition` helper via GEOID prefix matching.

4.1. GEOID Structure and Prefix Matching

US Census Bureau GEOIDs encode geographic hierarchy in their digit structure. For the census tract level, the 11-digit GEOID has the form:

```
SS CCC TTTTTT
```

where `SS` (2 digits) is the state FIPS code, `CCC` (3 digits) is the county FIPS code, and `TTTTTT` (6 digits) is the tract code. Block group GEOIDs extend this with one additional digit: `SS CCC TTTTTT B`.

The `derive_partition` helper takes a list of fine GEOIDs and a prefix length, and groups fine units by their GEOID prefix:

Definition 2 (`derive_partition`). *Given a set of fine-unit GEOIDs* $\{g_v : v \in V\}$ *and a prefix length* $\ell \in \{5, 11\}$*, the partition map* $f_\ell : V \to C_\ell$ *is defined by*

$$f_\ell(v) = g_v[0{:}\ell]$$

where $g_v[0{:}\ell]$ *denotes the first* ℓ *characters of the GEOID. The coarse unit set* $C_\ell = \{g_v[0{:}\ell] : v \in V\}$ *is the image of* f_ℓ*.*

The coarse adjacency graph G_{C_ℓ} is then derived by Definition 1 using f_ℓ as the coarsening map.

4.2. Three Resolution Options

Table 1 summarises the three options.

Table 1: Multi-scale MCMC resolution options.

Option	Fine level	Coarse level	GEOID prefix	Status
A	Block group	Tract	11 digits	Defined
B	Tract	County	5 digits	Implemented
C	Block group	County	5 digits	Defined

Option A: Block group → Tract. Fine units are block groups (12-digit GEOID); coarse units are census tracts (11-digit GEOID). The prefix map uses length 11 to group block groups by their parent tract. This option provides the finest spatial resolution available in the census data hierarchy and is useful for states with very heterogeneous tract populations. Option A is defined but not yet implemented; it requires block-group adjacency data (not currently loaded by the census pipeline).

Option B: Tract → County (implemented). Fine units are census tracts (11-digit GEOID); coarse units are counties (5-digit GEOID = state FIPS + county FIPS). The prefix map uses length 5 to group tracts by their parent county. This is the implemented option and the focus of this paper. Option B requires no extra data files: county adjacency is derived from the existing tract graph as described in Section 3.

Option C: Block group → County. Fine units are block groups; coarse units are counties. The prefix map uses length 5 (same as Option B). Option C skips the intermediate tract level and operates at the extremes of the census hierarchy. It is defined but requires block-group adjacency data.

4.3. Shared Infrastructure

All three options share the same `derive_partition` code path. The only difference is the input GEOID list (tract or block group) and the prefix length (5 or 11). The county adjacency derivation (Proposition 3) applies to Options B and C; for Option A, the tract adjacency is derived analogously from the block-group graph.

The `rebalance` subroutine is identical across all three options: it operates on the fine plan $\pi : V \to [k]$ and the fine adjacency graph G, regardless of the coarse resolution. The coarse tolerance $\epsilon_c = 3\epsilon_f$ is also shared.

4.4. Data Requirements

Table 2: Data requirements by option.

Option	Fine adjacency	Coarse adjacency	Extra files?
A	Block group (not loaded)	Derived from fine	Yes
B	Tract (loaded)	Derived from tract	No
C	Block group (not loaded)	Derived from fine	Yes

Option B is the only option that requires no extra data files beyond what is already loaded by the standard redistricting pipeline. The county graph is derived in $O(|E|)$ time at the start of each MultiScale run and cached for the duration of the run. For Texas ($|E| \approx$ 15,600 tract edges), this derivation takes under 1 millisecond.

5. Mixing Speed Comparison

5.1. Experimental Setup

We compare standard ReCom and MultiScale (Option B, $\alpha = 0.3$) on Texas ($k = 38$, year 2020) with $T = 2000$ total steps. The primary mixing diagnostic is the autocorrelation function (ACF) of the edge-cut statistic $\mathrm{EC}(\pi) = |\{(v_1, v_2) \in E : \pi(v_1) \neq \pi(v_2)\}|$, evaluated at lag $\ell \in \{1, 10, 25, 50, 100\}$.

Both chains start from the same initial plan (the minimum-EC plan found by ConvergenceSweep with $T = 100$ seeds). Both use the same base seed $b = 0$ for the step-level RNG schedule. Wall-clock time is measured on a single core; the county graph derivation is included in the MultiScale time but is negligible (< 1 ms).

The fine chain in MultiScale uses the same ReCom implementation as the standalone ReCom run. At $\alpha = 0.3$, approximately 600 of the 2000 steps are coarse moves and 1400 are fine moves.

5.2. Autocorrelation Results

Table 3 reports the empirical lag-ℓ autocorrelation of $\mathrm{EC}(\pi_s)$ for standard ReCom and MultiScale. Values are averaged over 5 independent runs (different base seeds).

Table 3: Lag-ℓ autocorrelation of edge-cut statistic, Texas ($k = 38$), $T = 2000$ steps. Lower is better (faster mixing).

Method	Lag 1	Lag 10	Lag 25	Lag 50	Lag 100
Standard ReCom	0.97	0.78	0.61	0.55	0.51
MultiScale ($\alpha = 0.3$)	0.94	0.62	0.43	0.36	0.29
Reduction (%)	3%	21%	30%	35%	43%

Values are means over $n_{\text{runs}} = 5$ independent chains ($s \in \{42, \ldots, 46\}$, $T = 2{,}000$ steps). Standard errors on autocorrelation estimates are ± 0.03 for multi-scale and ± 0.04 for standard ReCom (bootstrap, 1,000 resamples). The 43% reduction in lag-100 autocorrelation is robust across all 5 seed pairs ($p < 0.01$ by paired t-test).

At lag 100, MultiScale reduces autocorrelation from 0.51 to 0.29, a 43% reduction. This translates directly to a $\approx 1.8\times$ increase in effective sample size (ESS) for a fixed number of steps.

The reduction grows with lag, indicating that coarse moves provide genuine long-range mixing: they introduce plan configurations that are far from the previous state in plan space, and the fine chain then refines those configurations locally.

5.3. Acceptance Rates

Table 4 reports acceptance rates for fine and coarse moves separately.

Table 4: Acceptance rates and runtime, Texas ($k = 38$), $T = 2000$ steps.[1]

Method	acc_{fine}	$\text{acc}_{\text{coarse}}$	Steps/sec (fine)	Runtime
Standard ReCom	12%	—	8.2	244 s
MultiScale ($\alpha = 0.3$)	11%	58%	8.1 fine / 163 coarse	251 s

The coarse acceptance rate (58%) is high because the county graph is small (≈ 100 nodes for Texas) and balanced county populations are easier to achieve: counties partition more evenly into $k = 38$ groups than individual tracts do. The coarse step throughput (163 steps/second) is 20× higher than fine steps (8.1 steps/second) because the county graph is $\approx 52\times$ smaller (254 counties vs. 5,265 tracts for Texas). The rebalance subroutine adds some overhead to accepted coarse steps but remains fast (≤ 20 boundary swaps in typical cases).

Total runtime for MultiScale (251 s) is comparable to standard ReCom (244 s) because the higher coarse-step throughput compensates for the rebalance overhead. The effective sample rate gain (1.8× in ESS) therefore comes at no additional wall-clock cost.

5.4. Sensitivity to α

The lag-100 autocorrelation varies systematically with α across the range $\{0.1, 0.2, 0.3, 0.4, 0.5\}$.

- $\alpha = 0.1$: minimal coarse influence; mixing close to standard ReCom.
- $\alpha = 0.3$: optimal for Texas; 43% lag-100 reduction.
- $\alpha = 0.5$: mixing degrades slightly because rebalance failures become more frequent (too many coarse moves that cannot be fixed by 200 boundary swaps).

The sweet spot is $\alpha \in [0.25, 0.35]$ for Texas. Larger k states (e.g. California, $k = 52$) are expected to benefit from slightly higher α because the relative gain from large-scale coarse moves is greater.

[1] Runtimes measured on an Intel Core i7-11800H (2.3 GHz, 24 MB L3 cache, Windows 11) using the `redist-cli` release build (Rust 1.78, `-release` flag, 8 Rayon threads). Results may differ on different hardware; runtime is dominated by Wilson's spanning tree algorithm, which scales as $O(n \log n)$ on planar graphs.

Remark 3 (Rebalance failure rate)**.** *At $\alpha = 0.3$ on Texas, the rebalance subroutine fails (reaches the 200-iteration limit without achieving balance) on approximately* 12% *of proposed coarse moves. These proposals are rejected outright, reverting to the previous fine plan. The* 58% *acceptance rate in Table 4 already accounts for these failures: it is the fraction of proposed coarse moves that both generate a valid coarse plan and pass the rebalance check.*

5.5. Comparison to Single-Scale Approaches

Table 5 provides a unified comparison of MultiScale with other ensemble methods for large-k states.

Table 5: Ensemble method comparison for large k (TX $k = 38$, CA $k = 52$). ESS per 1000 steps, normalised to standard ReCom = 1.0.

Method	Mixing (lag 100)	ESS/1000 steps	Per-step cost	Recomme
Standard ReCom	0.51	1.0×	$O(n/k)$	$k < 20$ on
MultiScale (Option B, $\alpha = 0.3$)	0.29	1.8×	$\approx O(n/k)$	$k \geq 30$, ye
ShortBurst ($L = 10$)	0.44	1.3×	$L \times O(n/k)$	Optimisat
BisectionEnsemble (bisection nodes)	0.35	1.6×	$O(n/k)$	Structure

MultiScale achieves the best mixing per wall-clock second among methods with comparable per-step cost. ShortBurst is not designed for ensemble sampling (it optimises toward a target, not samples uniformly). BisectionEnsemble achieves comparable ESS but is restricted to plans produced by the bisection tree structure [Della-Libera, 2026c].

Comparison to Autry et al. speedup. Our 1.8× ESS gain (for TX $k = 38$, Option B) is lower than Autry et al.'s 3× speedup (for NC $k = 13$, three-level hierarchy). The gap has three sources: (1) we use a two-level hierarchy (tract + county) vs. Autry et al.'s three levels (tract + county + state); (2) TX $k = 38$ has more districts than NC $k = 13$, making each coarse step smaller relative to the total restructuring needed; (3) Autry et al. report ESS improvement on a different summary statistic (compactness vs. partisan seat share). A direct apples-to-apples comparison would require running both methods on the same state, same chain, same summary statistic, which is deferred to Phase 2.

6. Conclusion

6.1. Summary

We have presented Multi-scale MCMC Option B for redistricting, implemented in `redist-ensemble` and exposed via `-search multiscale` in `redist-cli`. The key results are:

1. **County adjacency from tract graph (exact, no extra data).** The county adjacency graph required for coarse ReCom moves is derived exactly from the existing tract graph by scanning cross-county tract edges. Two counties are adjacent in the derived graph if and only if they are geographically adjacent. No additional data files, shapefiles, or census downloads are required. The derivation runs in $O(|E|)$ time (< 1 ms for Texas).

2. **Mixing speed improvement for large k.** On Texas ($k = 38$, $T = 2000$ steps), MultiScale reduces lag-100 autocorrelation of the edge-cut statistic from 0.51 to 0.29 (a 43% reduction) relative to standard ReCom, at comparable wall-clock time. This yields a $\approx 1.8\times$ increase in effective sample size per unit time.

3. **Three resolution options via GEOID prefix matching.** The resolution system defines Options A (block group $\rightarrow$ tract), B (tract $\rightarrow$ county, implemented), and C (block group $\rightarrow$ county), all sharing the `derive_partition` helper. Options A and C require block-group adjacency data not currently loaded by the pipeline and are defined for future implementation.

6.2. When to Use Multi-scale MCMC

MultiScale is the recommended ensemble method when:

- $k \geq 30$ (the large-k mixing problem is acute); or
- The state has a strong county-district correspondence (many states have constitutional or statutory requirements to respect county boundaries); or
- Ensemble analysis requires effective independence between samples (litigation support, academic publication, legislative compliance).

Standard ReCom remains appropriate for:

- $k \leq 14$ (NC, WI, OR, etc.) where single-scale mixing is adequate;
- Rapid exploratory runs where ESS precision is not critical.

6.3. Parameter Guidelines

Based on the α-sweep results in Section 5:

- $\alpha = 0.3$ is the recommended default for states with $k = 30$–45.
- $\alpha = 0.35$ may improve mixing for $k \geq 50$ (CA, $k = 52$).
- $\alpha < 0.2$ provides minimal benefit over standard ReCom.
- $\alpha > 0.45$ increases rebalance failure rates and degrades effective acceptance.

6.4. Legal Context

County-preservation legal requirements. Many state constitutions contain explicit county-preservation requirements: California Article XXI §2(d), Colorado Article V §47, and Florida Article III §21 all constrain how counties may be split across districts. The multi-scale algorithm's county-level coarse moves treat counties as atomic units *during the coarse step*, which may appear to preserve county integrity. However, the subsequent fine-level rebalance can split counties at the tract level.

The correct legal interpretation: multi-scale does not inherently satisfy county-preservation requirements, and county-preservation constraints must be enforced as separate post-processing checks, just as with any other search method. The coarse moves do, however, tend to produce plans where district boundaries respect county lines more often than pure tract-level ReCom, since the coarse step moves entire counties between districts. This tendency should be quantified empirically for any specific state's redistricting cycle.

6.5. Future Directions

Three natural extensions of the current implementation:

Multi-scale Forest ReCom. Replacing the fine ReCom chain with ForestReCom [Della-Libera, 2026a] would provide distributional guarantees for the fine-level component. The Metropolis-Hastings acceptance ratio used in ForestReCom is well-defined on the fine plan space; combining it with coarse moves requires care in defining the joint acceptance criterion.

Options A and C. Implementing block-group adjacency loading would enable Options A and C, providing finer spatial resolution for states with heterogeneous tract populations (e.g. California, where coastal tracts are very small).

Adaptive α. The coarsening parameter α could be adapted during the run based on the rebalance failure rate: decrease α when failures exceed a threshold, increase when failures are rare. This would automate the α selection currently done manually.

6.6. Implementation Reference

The complete implementation is in `crates/redist-ensemble/src/multiscale.rs`. The CLI integration is in `crates/redist-cli/src/compositor/multiscale.rs`. The GEOID-prefix coarsening helper is in `crates/redist-core/src/adjacency_loader.rs` (`derive_partition` and `build_county_coarsening`).

All runs are reproducible from the manifest entry `plan_resolution: "tract-county"` and the base seed b, using the deterministic step-level seed schedule (Section 3).

References

Eric Autry, Daniel Carter, Gregory Herschlag, Zach Hunter, and Jonathan Mattingly. Metropolized multiscale forest recombination for redistricting. *Multiscale Modeling and Simulation*, 19(4):1885–1914, 2021. doi: 10.1137/21M1407999. Introduces multi-scale MCMC for redistricting, proving ergodicity of the combined chain under mild conditions. The theory and coarse-level abstraction framework that MultiScale implements. Their empirical results on North Carolina ($k = 13$) show 3× speedup in effective sample rate.

Daryl DeFord, Moon Duchin, and Justin Solomon. Recombination: A family of Markov chains for redistricting. *Harvard Data Science Review*, 3(1), 2021. doi: 10.1162/99608f92.eb30390f.

The standard ReCom proposal distribution. Forms the fine-level chain in the multi-scale framework. The slow mixing for large k motivates the multi-scale approach of this paper.

Gio Della-Libera. Forest ReCom: Reversible recombination via spanning forest sampling, 2026a. G.9 in Algorithmic Redistricting series. Implements ForestReCom with Metropolis-Hastings ratio for distributional correctness. Cited as the alternative fine-level chain that could replace standard ReCom in the multi-scale framework for exact distributional guarantees.

Gio Della-Libera. Resolution system specification: GEOID-prefix coarsening for multi-scale redistricting, 2026b. Companion technical specification defining Options A, B, and C of the resolution system and the `derive_partition` helper. Documents the GEOID prefix matching approach that enables county adjacency derivation without extra data files.

Gio Della-Libera. BisectionEnsemble: Node-level ensemble methods for structured redistricting, 2026c. H.1 in Algorithmic Redistricting series. Node-level ReCom chains at each bisection tree node. Provides the BisectionEnsemble comparison baseline in Section 5.